blurred when seen close up. Where exactly does the mountain end and where do the plains begin? Where does the savannah begin and the desert end? We cut the world into large slices. We think of it in terms of concepts that are meaningful for us, that *emerge* at a certain scale.

- We see the sky turning around us every day, but we are the ones who are turning. Is the daily spectacle of a revolving universe "illusory"? No, it is real, but it doesn't involve the cosmos alone. It involves *our relation* with the sun and the stars. We understand it by asking ourselves how *we* move. Cosmic movement *emerges* from the relation between the cosmos and ourselves.

In these examples, something that is real—a cat, a football team, high and low, the surface of clouds, the rotation of the cosmos—*emerges* from a world that at a much simpler level has no cats, teams, up or down, no surfaces of clouds, no revolving cosmos. . . . Time emerges from a world without time, in a way that has something in common with each of these examples.

The reconstruction of time begins here, in two little chapters—this one and the next—that are brief and technical. If you find them heavy going, skip them and go directly to chapter 11. From there, step by step, we will gradually reach more human things.

THERMAL TIME

In the frenzy of thermal molecular mingling, all the variables that can possibly vary do so continuously.

One, however, does not vary: the total amount of energy in any isolated system. Between energy and time there is a close bond. They form one of those characteristic couples of quantities that physicists call "conjugate," such as position and momentum, or orientation and angular momentum. The two terms of these couples are tied to each other. On the one hand, knowing what the energy of a system may be[78]—how it is linked to the other variables—is the same as knowing how time flows, because the equations of evolution in time follow from the form of its energy.[79] On the other, energy is conserved in time, hence it cannot vary, even when everything else varies. In its thermal agitation, a system[80] passes through all the configurations that have the same energy, but only these. The set of these configurations—which our blurred macroscopic vision does not distinguish—is the "(macroscopic) state of equilibrium": a placid glass of hot water.

The usual way of interpreting the relation between time and state of equilibrium is to think that time is something absolute and objective; energy governs the time-evolution of a system; and the system in equilibrium mixes all configurations of equal energy. The conventional logic for interpreting this relation is therefore: